Anurag K S V
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PROFESSIONAL EXPERIENCE

Wipro Limited, Wipro Innovation Network | Bengaluru, Karnataka, India (Hybrid, Full-Time) Jul 2026 – Present
Sr. Software Engineer (Quantum Research)

Qclairvoyance Quantum Labs | Secunderabad, Telangana, India (On-Site, Full-Time) Jul 2024 – Jul 2026
Quantum Analyst (Founding Team)

- Spearheading the design and development of a **proprietary quantum algorithm and software stack**, enabling modular hybrid workflows across NISQ and FASQ/FTQC paradigms.
- Directing technical pre-sales and stakeholder discovery, acting as **the primary technical bridge for academic partners, investors, and enterprise clients** to translate complex quantum software architectures into commercial workflows.
- Architecting the **quantum algorithmic layer** for hybrid computational chemistry pipelines for drug discovery, collaborating with cross-functional AI/ML (vHTS) and classical QM/MM teams to simulate Protein-Ligand interactions for complex **classically intractable problems**.
- Developing and **integrating quantum-classical hybrid solvers** with classical fragmentation and embedding methods to enable scalable molecular simulations on **IBM Quantum hardware** (current results on **32-40 qubits**) and **IQM Quantum Hardware** (current results on **16 qubits**) for industry-relevant drug molecules such as Amantadine (Mol. Wt. = 151 Da).
- Executed **1D and high-resolution 2D Potential Energy Surface (PES)** mapping for molecular systems (including complex grid evaluations for H_2O) on IQM Quantum hardware, benchmarking energy accuracies across STO-3G and 6-31G basis sets against exact reference solvers.
- Attained chemical-accuracy molecular simulations on IBM Quantum hardware (**4 to 20-qubits**), implementing advanced **error suppression and chemistry-aware mitigation frameworks** to validate algorithmic robustness under real-device noise.
- Prototyping **FTQC/FASQ subroutines** and conducting state-vector-based performance evaluations to guide resource estimation and circuit-depth optimization.
- Achieved up to **14× gate-resource reduction** and **11× simulation speed-up** for **12-qubit** molecular systems using **2,00,000 quantum gates** through **GPU** acceleration with **CUDA-Q**, scalabled to **32-qubit** architectures.
- Developed an in-house **quantum circuit resource estimation** tool using Qiskit and CUDAQ to evaluate ansatz expressibility, optimizer convergence, and Hamiltonian reduction strategies for molecular systems; extended it to handle general molecular configurations (orbitals, electrons) for resource estimation beyond **100 qubits**.
- Architecting **full-stack infrastructure** for quantum education and research platforms, integrating frontend, backend, and cloud compute pipelines for seamless user and developer workflows.
- **Surveying and benchmarking** the quantum computing hardware ecosystem: including **digital** platforms (IBM, IonQ, Quantinuum, Rigetti, IQM, OQC, PsiQuantum) and **analog** systems (D-Wave, Xanadu, Pasqal, QuEra), alongside error-correction methodologies and algorithmic performance metrics at qubit, system, and application levels to inform technology roadmaps.

Hexawel Healthcare | Hyderabad, Telangana, India (Remote, Part-Time) Jun 2021 – Jul 2022
Business Data Analyst

- Designed and managed an operations management system, enabling seamless end-to-end product tracking and significantly reducing paper-based and labor-intensive processes.
- Developed an interactive Google Looker Studio dashboard for visualizing purchases, inventory, and sales forecasts, empowering stakeholders with real-time insights for informed decision-making.

TECHNICAL SKILLS

Quantum Computing: Qiskit, Cirq, PennyLane, CUDA-Q, Tket, QARP
Accelerated Computing: NumPy, Pandas, Numba, CuPy, OpenMP, Nvidia CUDA
Computational Quantum Chemistry: PySCF, Psi4, Gaussian, Amber Tools, GROMACS
Core Programming: C, C++, Python, Java, Structured Query Language (SQL)
Backend & AI Integration: Flask, FastAPI, Celery, RabbitMQ, PostgreSQL, MongoDB, LLM APIs (Groq/Llama), Local (Ollama)
Business Intelligence: Google Looker Studio, Power BI, Matplotlib, Seaborn, Plotly
Hardware Description & Simulation: VHDL, System Verilog, AMD Xilinx Vivado
Cloud & DevOps: Docker, Docker Compose, Google Cloud Platform (GCP), Amazon Web Services (AWS)
Operating Systems: Linux (Ubuntu, CentOS), Windows Subsystem for Linux 2 (WSL2), Microsoft Windows
IDEs & Development Tools: Git, GitHub, Bash, LATEX, Visual Studio Code, Cursor, Windsurf, IntelliJ, Anaconda Distribution Suite

EDUCATION

Defence Institute of Advanced Technology (DIAT) | Pune, Maharashtra, India — **May 2024**
Master of Technology: Quantum Computing (Specialization in Quantum Communication & Sensing) — GPA: 9.05/10.0
Gandhi Institute of Technology and Management (GITAM) | Visakhapatnam, Andhra Pradesh, India — **May 2022**
Bachelor of Technology: Computer Science and Engineering — GPA: 9.06/10.0

PUBLICATIONS

- Ashish K. P., **Anurag K. S. V.**, Sai S. P. *et al.* Quantum Simulation of Ligand-like Molecules through Sample-based Quantum Diagonalization in Density Matrix Embedding Framework. *arXiv* (Nov 2025) DOI: 10.48550/arXiv.2511.22158 [accepted: *Journal of Chemical Physics*] (Oct 2026)
- **Anurag K. S. V.**, Ashish K. P., Vikas D. G. *et al.* Resource Estimation for VQE on Small Molecules: Impact of Fermion Mappings and Hamiltonian Reductions. *Journal of Computational Chemistry* (Apr 2026) DOI: 10.1002/jcc.70379 [arXiv: 2512.01605]
- **Anurag K. S. V.**, Shubham C., Srinivasan K. *et al.* Engineering a Phase-Noise-Based Quantum Random Number Generator for Real-Time Secure Applications: Design, Validation, and Scalability. *Proc. SPIE 14168, Sixth International Conference on Optical and Wireless Technologies (OWT 2025)*, 141682I (Mar 2026) DOI: 10.1117/12.3108280 [arXiv: 2604.00741]
- Shubham C.[†], **Anurag K. S. V.**[†], Raghavan G. *et al.* FPGA-based Toeplitz Strong Extractor for Quantum Random Number Generators. *2024 IEEE 5th India Council International Subsections Conference (INDISCON)*, pp. 1-5. (Aug 2024) DOI: 10.1109/INDISCON62179.2024.10744392 ([†]contributed equally to the work) [arXiv: 2505.02868]
- **Anurag K. S. V.**, Raghavan G., & Kanaka Raju P. High-efficiency Implementation of Toeplitz Strong Extractor for PRNG and QRNG output on CPU/GPU hardware systems. *Physica Scripta,* 99(7), 075115. (Jun 2024) DOI: 10.1088/1402-4896/ad587f
- **Anurag K. S. V.,** & Bhaskara Rao B. Temporal and Correlational Analysis of Air Pollution and Covid-19 Across Major Metropolis in India*. International Journal of Innovative Technology and Exploring Engineering,* 11(11), 1-6. (Oct 2022) DOI: 10.35940/ijitee.K9277.10111122

PREPRINTS

- **Anurag K. S. V.**, Ashish K. P., Afra N. *et al*. Application-Oriented Evaluation of CPU/GPU Quantum Simulators: From Foundational Circuits to Quantum Chemistry. *<under-review>* (Oct 2026)
- Mousumi K., Ashish K. P., **Anurag K. S. V.** *et al*. An Analytically Trained Variational Surrogate for Quantum Phase Estimation on NISQ Hardware. *arXiv* (Jul 2026) DOI: 10.48550/arXiv.2607.20943
- Ashish K. P., **Anurag K. S. V.**, Ruchika B. *et al.* Machine-Learned Compact Subspace Generation for Quantum Selected Configuration Interaction within Density Matrix Embedding Framework. *arXiv* (Jul 2026) DOI: 10.48550/arXiv.2607.20585
- **Anurag K. S. V.**[†], Ashish K. P.[†], Manas M. *et al.* Bridging the NISQ and Fault-Tolerant Regimes: Generative-ML-Assisted Quantum Selected CI for Molecular Simulations. arXiv (May 2026) <under-review> ([†]contributed equally to the work)
- **Anurag K. S. V.**[†], Ashish K. P.[†], Manas M. *et al.* Towards Chemically Accurate and Scalable Quantum Simulations on IQM Quantum Hardware: A Quantum-HPC Hybrid Approach. *arXiv* (Apr 2026) DOI: 10.48550/arXiv.2604.01983 ([†]contributed equally to the work)
- Akash K. S., Ashish K. P., **Anurag K. S. V.** *et al*. A Polylogarithmic-Time Quantum Algorithm for the Laplace Transform. *arXiv* (Dec 2025) DOI: 10.48550/arXiv.2512.17980
- Ashish K. P., **Anurag K. S. V.**, Vikas D. G. *et al.* Survey of Quantum Algorithms: Foundations, Frameworks and Applications. *Authorea* (Nov 2025) DOI: 10.22541/au.176341037.72812424/v1
- **Anurag K. S. V.**, Ashish K. P., Chinmay A. *et al.* Potential Energy Surface Scan of n-butane using Various Quantum Chemistry Software. *Authorea* (Jul 2025) DOI: 10.22541/au.174624847.78353339/v2

INVITED PANELS & LECTURES

- **Expert Session,** "Theory and Implementation of VQE for Molecular Systems" Delivered an expert session during the Quantum Computing Bootcamp 2026, conducted by Qclairvoyance Quantum Labs in collaboration with SVNIT Surat, (May 2026).
- **Invited Speaker,** "The Quantum Code: Mastering Algorithms." Delivered an alumni expert upskilling session hosted by the Training and Competency Development GITAM Career Guidance Cell, (Mar 2026).
- **Panel Member,** "Careers in Quantum." Represented Qclairvoyance Quantum Labs at Qiskit Fall Fest 2024 (QuantumVerse Vol 2). Hosted by the IEEE Student Branch at BITS Pilani Hyderabad Campus (Nov 2024).

MASTER'S THESIS

Anurag K. S. V., Development of a High-Speed Quantum Random Number Generator with Post-Processing Techniques for Randomness Extraction. M.Tech Thesis, *Defence Institute of Advanced Technology, Pune* (May 2024)
Advisors: Kanaka Raju P., & Raghavan G.

- **Physical System:** Developed a modular QRNG using delayed self-heterodyne phase-fluctuations measurement with Gbps-rate raw data generation.

- **Post Processing:** Implemented randomness extraction using FPGA (AMD Xilinx) and FFT based GPU-accelerated pipelines on high-performance systems (Nvidia V100 GPUs on CDAC Param Shavak).
- **Randomness Tests:** Extracted data successfully passed NIST SP 800-22, Diehard test suites for statistical randomness.
- **Speed:** Achieved 19.5 Gbps extraction-rate suitable for Quantum Key Distribution (QKD) applications.
- **Productization:** Packaged QRNG with FPGA based post-processing prototype reached TRL 7, approaching 8.

POSTER PRESENTATIONS

- **Anurag K. S. V.**, Kanaka Raju P., & Raghavan G. Development of High-Speed Quantum Random Number Generator harvesting Quantum Phase Fluctuations. *International Conference on Quantum Technologies and Applications 2024 (ICQTA 2024)* (Feb 2024) DOI: 10.13140/RG.2.2.22320.98569
- **Anurag K. S. V.**, Raghavan G., & Kanaka Raju P. Implementation and Evaluation of Toeplitz Strong Extractor for Post Processing of QRNGs on various Hardware Systems. *International Conference on Quantum Technologies and Applications 2024 (ICQTA 2024)* (Feb 2024) DOI: 10.13140/RG.2.2.33226.17606

CONFERENCES

• OWT 2025 | NIT Goa	Goa, India	Oral Presenter	Dec 2025
• QT4HEP | CERN	Geneva, Switzerland	Virtual attendee	Jan 2025
• ICQTA 24 | MAHE	Manipal, India	Poster Presenter	Feb 2024
• Quantum 2.0 | Optica (Formerly OSA)	Denver, CO, USA	Virtual Attendee	Jun 2023

LABORATORY EXPERIENCE

Lasers & Optics: Hands-on exposure to using optical components such as lasers, mirrors, lenses, waveplates, beam splitters, and Acousto-Optic Modulators (AOMs) to perform fundamental optics-based experiments including free-space, fibre-based interferometry, stokes parameter measurement for polarization of light.

Quantum Communication: Performed experiments on quantum entanglement generation employing the Spontaneous Parametric Down Conversion (SPDC) phenomenon realized using a Beta-Barium Borate (BBO) Crystal and its characterization via quantum state tomography, Clauser-Horne-Shimony-Holt (CHSH) parameter and the second-order correlation function. Worked on the prototypical demonstration of the B92 Quantum Key Distribution (QKD) Protocol.

Light Matter Interaction: Worked on experiments including observation of Saturation Absorption Spectroscopy (SAS) while passing laser light through atomic (Rb) vapour cell.

Electronics: Worked with various photodetectors including High Bandwidth InGaAs, Avalanche, Single Photon Avalanche Diodes (SPAD), High Speed Oscilloscopes, Radio-Frequency (RF) Spectrum Analyzers, and Optical Spectrum Analyzer (OSA) Systems.

CERTIFICATIONS & WORKSHOPS

IQM Quantum Computers | Certificate of Participation – IQM Quantum School 2025 (Online)	Dec 2025
IBM Quantum | Badge of Excellence & Certificate of Participation – Qiskit Global Summer School 2024 (Online)	Aug 2024
IIT Bombay | Certificate of Participation – ICONS2023 Two-Day Workshop on Quantum Technologies (Mumbai, India)	Nov 2023
CDAC India | Certification of Completion with Merit – Quantum Computing using QSim (Online)	Jul 2023
SETS Chennai | Certificate of Training – FPGA based post-processing for QKD systems (Chennai, India)	Jun 2023
IBM Quantum | Quantum Spring 2023 Challenge	Apr 2023

LEADERSHIP & COMMUNITY ENGAGEMENT

Directorate of Sports & NCC GITAM Jul 2020 – Jun 2022

Student Sports Associate (SSA)

- Founded and institutionalized the GITAM Sports Committee (GSC), coordinating 30 student representatives across three campuses (Visakhapatnam, Hyderabad, Bengaluru).
- Led execution of 50+ online and hybrid engagement events during the COVID-19 period to sustain student participation.
- Supported the Directorate in drafting governance frameworks, including event policies, guidelines, and recruitment procedures for Sports Interest Groups (SIGs).
- Oversaw the creation and operations of multiple SIGs, scaling participation to 1,750+ active student members across campuses.
- Mentored SIG representatives in team leadership, operational planning, and on-ground execution for training programs, tournaments, and leagues.
- Planned and coordinated flagship university sports events, including GITAM Sports Week and the inaugural “Sangraam” Inter-Residential Championship (Visakhapatnam campus).

HONORS, AWARDS & ACHIEVEMENTS

- Received **Best Paper Award** for the session at the 6th International Conference on Optical & Wireless Technologies (OWT 2025).
- Secured **3rd position** in the IBM Qiskit Fall Fest 2023 QUantest: the coding challenge organized by IBM Quantum and DIAT.
- Secured **85th percentile** in Graduate Aptitude Test in Engineering (Physics) 2022 with All India Rank - 2879.
- Graduated in the **95th percentile** in a cohort of 1080 students, receiving a Merit Scholarship for semester 4 (2019 top 5% of CSE Dept.) during my undergraduate studies.

Languages:

English (Fluent-C1), Telugu (Native-C2), Hindi (Fluent-C1), Japanese (Basic-A1).